The Dan Campbell Effect

Fourth Down Decision-Making in the National Football League

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Motivation

- League rules are designed to foster consistent parity among NFL franchises
- Teams have embraced on-field analytics in an effort to gain a competitive advantage
- Fourth downs can be the most consequential plays in a game in terms of impacting expected win probability

What is a fourth down?

- Fourth down is the final opportunity for an offense to continue their drive towards the opponent's end zone
- Traditional strategy has leaned towards conservative decision-making, essentially treating third down as the last chance
- Modern thinking is to maximize win expectancy, which often means extending a drive to score a touchdown

Research Gap

- Dan Campbell became the head coach of the Detroit Lions in 2021, implementing aggressive fourth down tactics
- Successfully conversions are praised, while failed ones are criticized as reckless
- Is the media narrative that Campbell changed fourth down strategy leaguewide true?

Data

- Play-by-play data was sourced from nflverse using `nflfastR`, supplemented with `nflplotR` for enhanced visuals

- Completed fourth down plays from the last 25 seasons
    - Over 90,000 rows
    - 16 numerical, 4 indicator, and 1 categorical columns

- For performance reasons, more complex models utilized the last 10 seasons of play-by-play data

Bayesian Methods

- STAN was used to fit a series of GLMMs via the `brms` package in R, using the Hamiltonian Monte Carlo algorithm

- **Model 1:** Logistic regression with beta likelihood (fixed)
    - Priors: Flat, Gaussian, gamma (sd)

- **Model 2:** Logistic regression with Bernoulli likelihood (mixed)
    - Priors: Flat, Gaussian, horseshoe (shrinkage), gamma (sd)

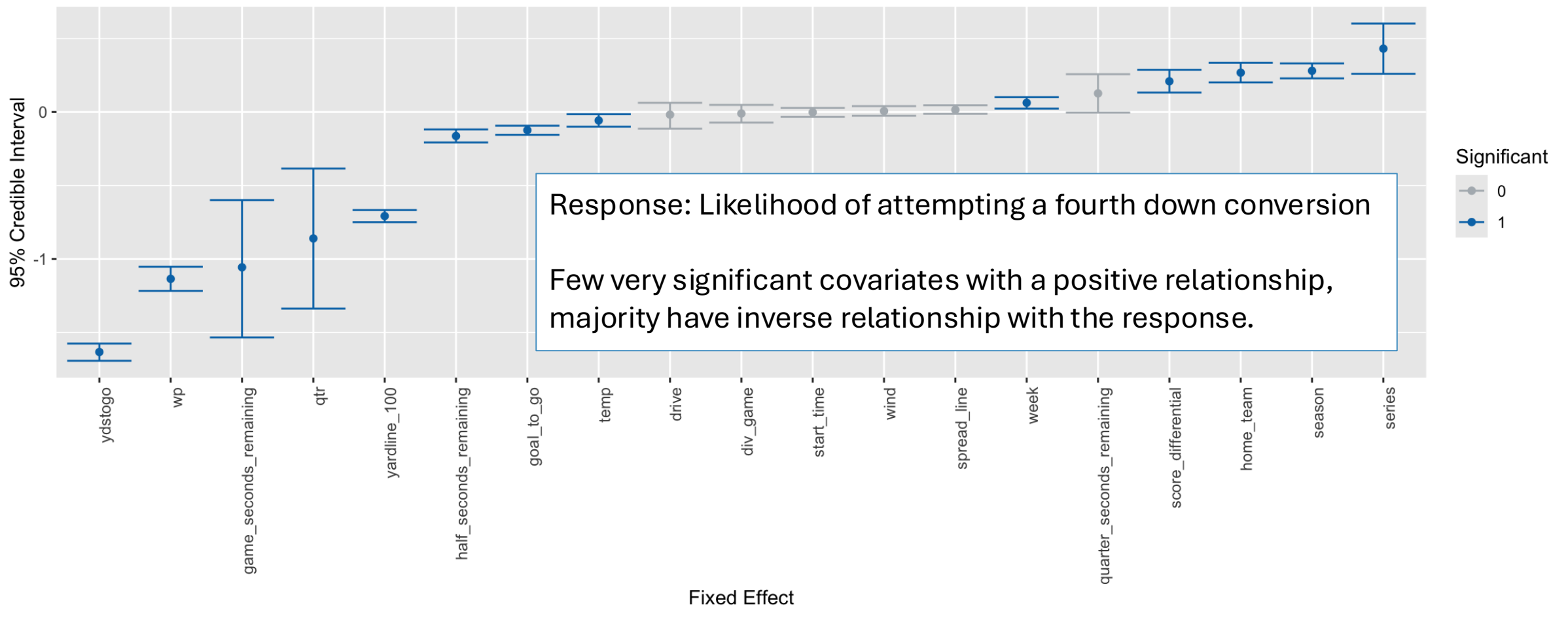
M1 Results: The Dan Campbell Effect

- The Campbell covariate is associated with a 0.367 increase in the log-odds of a conversion attempt proportion in an NFL season

- While the results were moderately sensitive to prior, Campbell always remained a significant predictor

Fixed Effect	Estimate	Error	Q2.5	Q97.5
Intercept	-1.87	0.0454	-1.95	-1.79
Season	0.103	0.0483	0.00798	0.199
Dan Campbell	0.367	0.109	0.155	0.582

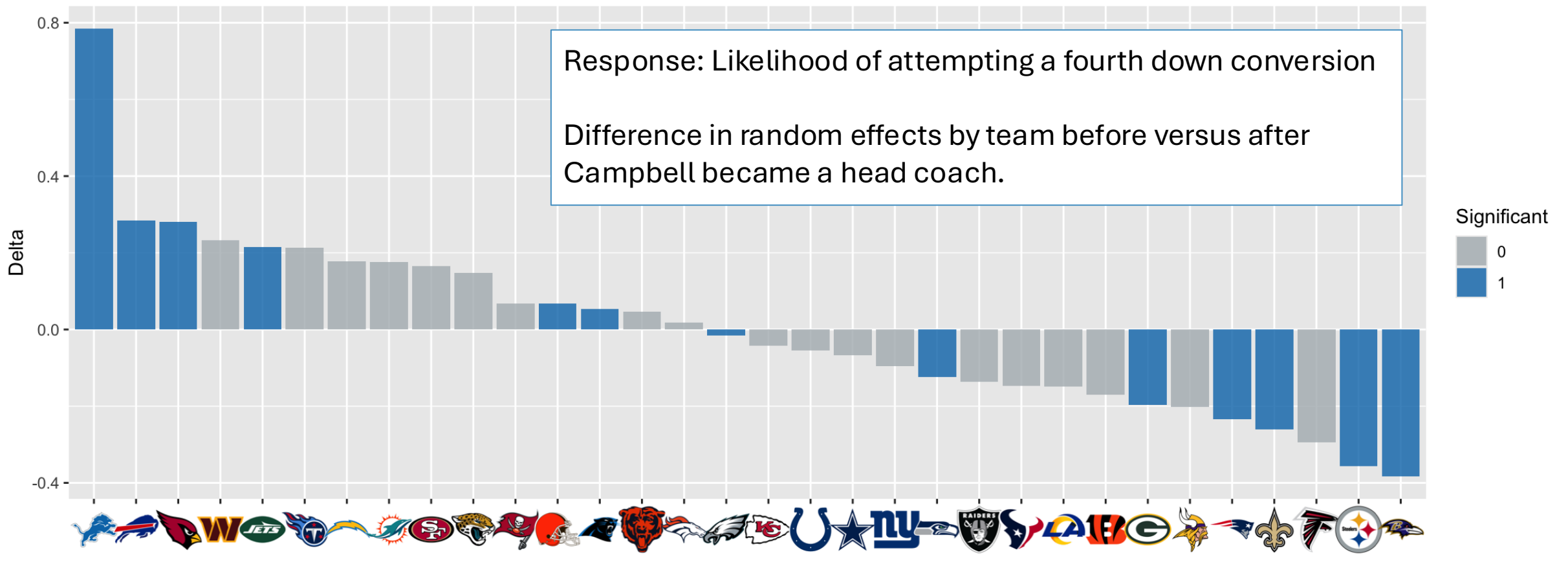
M2 Results: Fixed Effects



Response: Likelihood of attempting a fourth down conversion

Few very significant covariates with a positive relationship, majority have inverse relationship with the response.

M2 Results: Random Effect Delta



Response: Likelihood of attempting a fourth down conversion

Difference in random effects by team before versus after Campbell became a head coach.

Key Findings

- Strong evidence that conversion attempts are increasing over time
- Dan Campbell is associated with an increase in the proportion of fourth down conversion attempts since 2021
    - Impact is not universal, more pronounced for some teams
- Traditional fourth down strategy is still reinforced by significance of variables such as yards to go, goal to go, and win probability
- Factors like wind not nearly as significant as one might assume

Strengths and Opportunities

+ All models were fit on a significant amount of data, which lessens the impact of prior information

- Computational performance limited the number of model fits that could be efficiently ran

- As a Detroit Lions fan, I have a clear bias towards this research and could potentially be overstating significance of my findings

- Would like to extend this analysis to also investigate fourth down conversion success rate with additional models

References

Bürkner, P.-C. (n.d.). *brms: Bayesian regression models using Stan*. Retrieved April 30, 2026, from https://paulbuerkner.com/brms/index.html

nflfastR. (n.d.). *nflfastR: Fast and tidy NFL data*. Retrieved April 30, 2026, from https://nflfastr.com/index.html

Stan Development Team. (n.d.). *Stan documentation*. Retrieved April 30, 2026, from https://mc-stan.org/docs/

Wikipedia contributors. (2025). *Beta regression*. Retrieved April 30, 2026, from https://en.wikipedia.org/wiki/Beta_regression

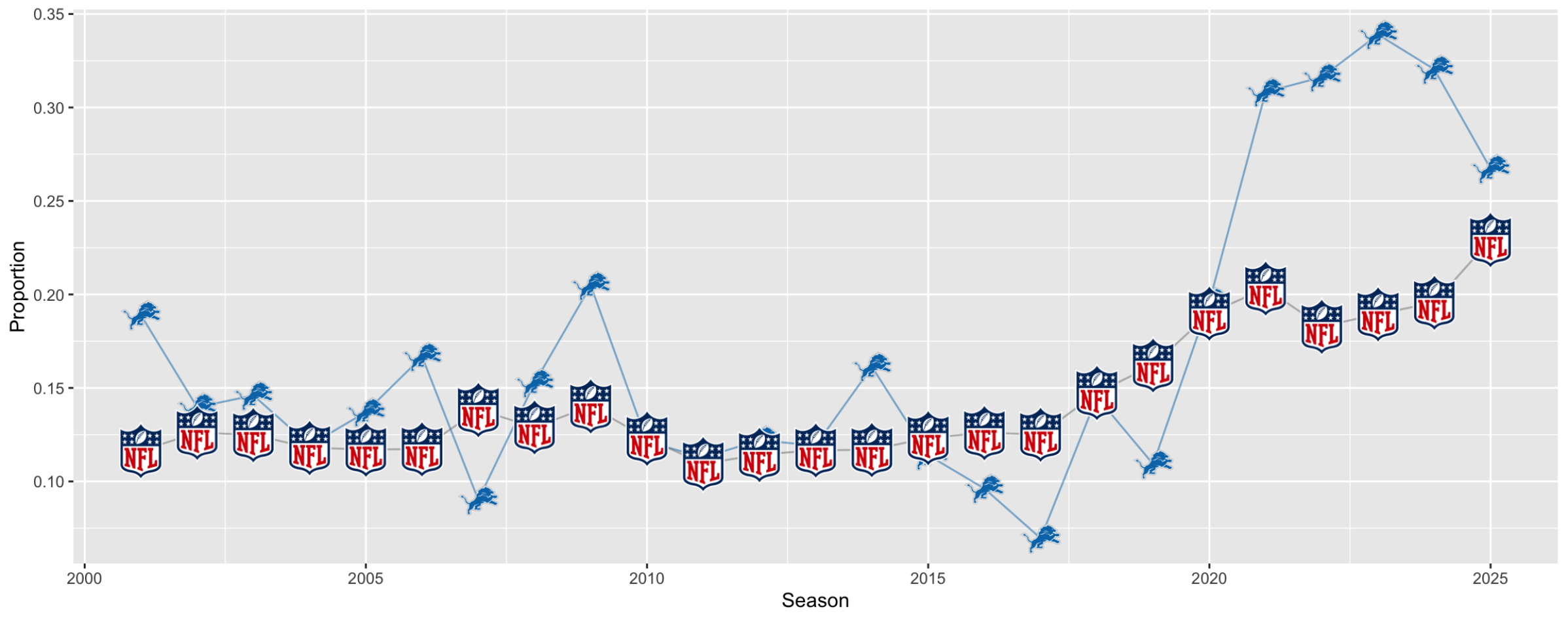
Models

Model 1: Fourth down conversion attempt proportion by season

$$\text{attempt_prop}_t \sim \text{Beta}(\mu_t\phi, (1 - \mu_t)\phi)$$
$$\text{logit}(\mu_t) = \beta_0 + \beta_1 \cdot \text{season} + \beta_2 \cdot \text{dan_campbell}$$

Model 2: Fourth down conversion attempt likelihood

$$\text{attempt}_t \sim \text{Bernoulli}(p_i)$$
$$\text{logit}(p_i) = \beta_0 + \sum_{j \in J} \beta_j \cdot \text{X}_j + \sum_{d \in D} \sum_{k \in K} \alpha_{dk}$$

J = fixed effects, D = binary indicator for Dan Campbell, K = NFL team

Proportion of Conversion Attempts by Season